

Modeling and Backstepping Stabilization of a Canted-Axis Gimbal on a 6-DoF Moving Base

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Abstract—This paper presents a robust nonlinear control framework for the stabilization of a 3-axis gimbal system featuring a non-orthogonal canted axis, mounted on a 6-DoF moving base. Addressing the limitations of traditional linear controllers in handling coupled nonlinear dynamics and kinematic singularities, a comprehensive mathematical model is derived using the Euler-Lagrange formulation. A hierarchical control architecture is proposed, integrating a Lyapunov-based Backstepping controller for precise motion tracking with a discrete-event supervisor to manage system safety and operational modes. Extensive simulations demonstrate the controller's effectiveness in rejecting external base disturbances and ensuring global asymptotic stability. Furthermore, the feasibility of the proposed approach is validated through the development of a custom experimental testbed, utilizing a distributed computing architecture comprising a host processor and embedded low-level drivers.

Index Terms—Gimbal stabilization, Backstepping control, Nonlinear dynamics, Euler-Lagrange modeling, Discrete-event supervisor.

I. INTRODUCTION

A 3-axis gimbal is a pivotal mechanism designed to isolate a payload, such as a camera or sensor, from the rotational motion of its base. This technology is critical in applications like aerial photography, satellite tracking, and robotics, where maintaining a stable orientation is paramount. The control of such a system is a challenging problem in mechatronics and control theory due to its inherently complex, coupled, and nonlinear dynamics.

Simple linear controllers such as PID can provide basic stabilization, but often fail to deliver high performance under aggressive maneuvers or in the presence of external disturbances [1]. This project aims to address these limitations by designing and implementing a sophisticated nonlinear controller to achieve superior stability and tracking performance.

II. PROBLEM FORMULATION

The primary objective of this project is to maintain the orientation of the gimbal's end-effector at a desired setpoint or trajectory, regardless of disturbances. This task presents a

multi-faceted challenge, spanning both mechanical design and advanced control theory.

A. Mechanical & Kinematic Complexity

Although many academic studies simplify the stabilization problem using 2-axis gimbals [2], this project tackles the full 3-axis system. The design and construction of a 3D gimbal are significantly more complex, introducing challenges such as link balancing, and avoiding kinematic singularities like "gimbal lock," a state where one degree of rotational freedom is lost. These physical complexities underscore the need for a controller that is not only theoretically sound but also robust to real-world mechanical imperfections.

B. The Nonlinear Model & Limitations of Linear Control

The dynamics of a 3-axis gimbal is described by a set of coupled, nonlinear differential equations. This nonlinearity arises from trigonometric terms in the Euler angle kinematics and the presence of state-dependent Coriolis and centrifugal forces. The resulting model is highly complex and its behavior changes drastically depending on the gimbal's orientation and angular velocity.

This inherent nonlinearity poses a significant problem for traditional linear controllers like PID. A PID controller is tuned for optimal performance around a single, specific operating point [3-4]. When the system undergoes aggressive maneuvers or is subjected to large disturbances, it moves far from this point, and the PID controller's performance degrades severely, often leading to sluggish response, overshoot, or even instability. It fundamentally lacks the structure to systematically compensate for known nonlinearities or adapt to parametric uncertainties, such as a change in payload.

C. Advanced Nonlinear Control

To overcome these limitations, advanced nonlinear control strategies are necessary. Two prominent candidates for this task are Backstepping and Sliding Mode Control (SMC), each

with a distinct set of advantages and disadvantages for this application.

One prominent strategy is Backstepping Control, which offers a systematic, recursive design process founded on Lyapunov stability [5]. This method provides a mathematical guarantee of stability for the known nonlinear system and allows for the precise cancellation of complex dynamic terms. However, it is not without drawbacks. Backstepping often suffers from an "explosion of complexity," resulting in algebraically massive control laws. More critically, its performance is highly dependent on perfect model knowledge, making it sensitive to parametric uncertainties (like incorrect inertia values) and unmodeled dynamics, which can lead to significant performance degradation in real-world conditions.

Another powerful alternative is Sliding Mode Control (SMC), whose defining feature is its exceptional robustness [6]. SMC is specifically designed to perform reliably even in the presence of significant parametric uncertainties and external disturbances, making it an ideal choice for physical systems where the mathematical model is never perfect. The primary disadvantage of standard SMC is a phenomenon known as "chattering." This involves high-frequency switching of the control signal, which can excite unmodeled dynamics and potentially cause mechanical wear on the system's motors over time.

The goal is therefore to design a control law that not only stabilizes the system but does so robustly, leveraging the strengths of these advanced methods while mitigating their weaknesses.

III. MATHEMATICAL MODEL

A. System Overview

The system consists of a three-axis gimbal mechanism mounted on a moving base platform. The system dynamics are modeled using the Euler-Lagrange formulation, treating the assembly as a multi-body system with 9 Degrees of Freedom (DoF).

The configuration of the system is described by a generalized coordinate vector $\mathbf{q} \in \mathbb{R}^9$:

$$\mathbf{q} = [x, y, z, \phi, \theta, \psi, \theta_r, \theta_p, \theta_y]^T \quad (1)$$

where the first six elements represent the base platform's linear position (x, y, z) and Euler orientation angles (ϕ, θ, ψ) . The last three elements $(\theta_r, \theta_p, \theta_y)$ represent the angular positions of the gimbal's BLDC motors.

Consequently, the full system state is represented by 18 state variables comprising the generalized coordinates and their time derivatives: $\mathbf{x} = [\mathbf{q}^T, \dot{\mathbf{q}}^T]^T$.

The system is driven by a generalized input vector $\boldsymbol{\tau} \in \mathbb{R}^9$:

$$\boldsymbol{\tau} = [\tau_{bx}, \tau_{by}, \tau_{bz}, \tau_{b\phi}, \tau_{b\theta}, \tau_{b\psi}, \tau_{m1}, \tau_{m2}, \tau_{m3}]^T \quad (2)$$

This input vector accounts for the six external forces and torques acting on the base degrees of freedom, alongside the three control torques applied by the gimbal actuators.

B. Dynamic Modeling

Following the system overview, the complete dynamic model is derived using the Euler-Lagrange formulation. The system is modeled as a serial kinematic chain comprising four rigid bodies: the moving base platform and the three articulated gimbal links. We define coordinate frames attached to the Center of Mass (CoM) of each body. The kinematic configuration is described using homogeneous transformation matrices, denoted as $T \in \mathbb{R}^{4 \times 4}$, which transform vectors from local body frames to the inertial frame.

1) *Base Transformation (T_{base})*: The base platform configuration is defined by the translation vector $\mathbf{p} = [x, y, z]^T$ and the orientation $\boldsymbol{\Theta} = [\phi, \theta, \psi]^T$. The transformation matrix T_{base} is constructed by applying the linear translation T_{pos} followed by three elemental rotations corresponding to the Euler roll (R_x), pitch (R_y), and yaw (R_z) sequences.

$$T_{base} = T_{pos} \cdot R_x(\phi) \cdot R_y(\theta) \cdot R_z(\psi) \quad (3)$$

Where:

$$T_{pos} = \begin{bmatrix} 1 & 0 & 0 & x \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) & 0 \\ 0 & \sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_z(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 & 0 \\ \sin(\psi) & \cos(\psi) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2) *Link 1 Transformation (T_{com1})*: The first link is attached to the base and actuated by the roll motor. Its CoM frame is derived by transforming the base frame by the roll joint rotation $R_y(\theta_r)$ and the rigid offset T_{off1} .

$$T_{com1} = T_{base} \cdot R_y(\theta_r) \cdot T_{off1} \quad (4)$$

Where:

$$R_y(\theta_r) = \begin{bmatrix} \cos(\theta_r) & 0 & \sin(\theta_r) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta_r) & 0 & \cos(\theta_r) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{off1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 1 & z_{c1} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3) *Link 2 Transformation (T_{com2})*: A critical feature of this system's geometry is the structural tilt angle, α . Unlike standard orthogonal gimbals, this system includes a fixed rotation of $\alpha = 15^\circ$ about the local x -axis between the roll and pitch joints. This geometric deviation is captured in the matrix T_{tilt} and creates a unique kinematic coupling between axes. The transformation T_{com2} accounts for the link length l_2 , the structural tilt, the pitch joint rotation $R_z(\theta_p)$, and the CoM offset T_{off2} .

$$T_{com2} = T_{link1} \cdot T_{disp} \cdot T_{tilt} \cdot R_z(\theta_p) \cdot T_{off2} \quad (5)$$

Where:

$$\begin{aligned} T_{disp} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ T_{tilt} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) & 0 \\ 0 & \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ R_z(\theta_p) &= \begin{bmatrix} \cos(\theta_p) & -\sin(\theta_p) & 0 & 0 \\ \sin(\theta_p) & \cos(\theta_p) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ T_{off2} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & y_{c2} \cos(\alpha) + z_{c2} \sin(\alpha) \\ 0 & 0 & 1 & -y_{c2} \sin(\alpha) + z_{c2} \cos(\alpha) \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

4) *Link 3 Transformation (T_{com3})*: The final link completes the chain. The transformation T_{com3} includes the translation to the yaw joint, an "un-tilt" operation T_{untilt} to realign the frame, and the final yaw rotation $R_y(\theta_y)$.

$$T_{com3} = T_{link2} \cdot T_{disp23} \cdot T_{untilt} \cdot R_y(\theta_y) \quad (6)$$

Where:

$$\begin{aligned} T_{disp23} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & y_{23} \cos(\alpha) + z_{23} \sin(\alpha) \\ 0 & 0 & 1 & -y_{23} \sin(\alpha) + z_{23} \cos(\alpha) \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ T_{untilt} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha) & \sin(\alpha) & 0 \\ 0 & -\sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ R_y(\theta_y) &= \begin{bmatrix} \cos(\theta_y) & 0 & \sin(\theta_y) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta_y) & 0 & \cos(\theta_y) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

The specific geometric parameters and offsets used in these transformations are summarized in Table I below.

TABLE I
GEOMETRIC PARAMETERS OF THE GIMBAL SYSTEM.

Symbol	Value	Description
l_1	0.08822 m	Link 1 Length
l_2	0.13548 m	Link 2 Length
α	15°	Structural Tilt Angle
z_{c1}	0.02554 m	Link 1 COM Z-offset
y_{c2}	-0.00051 m	Link 2 COM Y-local
z_{c2}	-0.06656 m	Link 2 COM Z-local
y_{23}	0.03734 m	Joint 2-3 Y-offset
z_{23}	-0.09309 m	Joint 2-3 Z-offset

5) *Dynamic Equations*: The system dynamics are formulated using the Euler-Lagrange method by extracting the energy properties directly from the kinematic definitions above.

For each body i in the chain, the linear velocity $\mathbf{v}_i$ of the Center of Mass is obtained by differentiating the position vector $\mathbf{p}_i$, which corresponds to the fourth column of the transformation matrix $T_{com,i}(\mathbf{q})$.

$$\mathbf{v}_i = \frac{d}{dt} (T_{com,i}[0..2, 3]) \quad (7)$$

The angular velocities $\boldsymbol{\omega}_i$ are computed recursively, summing the angular velocity of the predecessor link with the current joint rate, projected into the local body frame to align with the constant inertia tensors.

The Lagrangian $\mathcal{L}$ is defined as $\mathcal{L} = \mathcal{K} - \mathcal{P}$. The total kinetic energy ($\mathcal{K}$) is the sum of translational and rotational energies of the base and all three links:

$$\mathcal{K} = \sum_{i \in \{b, 1, 2, 3\}} \left(\frac{1}{2} m_i \mathbf{v}_i^T \mathbf{v}_i + \frac{1}{2} \boldsymbol{\omega}_i^T \mathbf{I}_i \boldsymbol{\omega}_i \right) \quad (8)$$

The potential energy ($\mathcal{P}$) is derived from the gravitational height of each CoM, extracted from the z -component of the transformation matrices:

$$\mathcal{P} = \sum_{i \in \{b, 1, 2, 3\}} m_i g h_i(\mathbf{q}) \quad (9)$$

Applying the Lagrange equation yields the equations of motion in standard matrix form:

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} \quad (10)$$

where $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{G}$ represent the inertia matrix, Coriolis/centrifugal matrix, and gravity vector, respectively.

C. Inverse Kinematics

The objective of the inverse kinematics module is to compute the required joint angles $\mathbf{q}_m = [\theta_r, \theta_p, \theta_y]^T$ that align the gimbal end-effector with a desired target orientation, given the instantaneous orientation of the base platform. Let $\mathbf{q}_{base}$ and $\mathbf{q}_{tgt}$ denote the quaternions representing the orientation of the base and the target, respectively.

1. *Orientation Alignment*: The solution begins by converting the input quaternions into rotation matrices R_{base}^{raw} and R_{tgt} . To account for the specific mounting interface of the gimbal, a coordinate transformation is applied to the base matrix by permuting its rows. This yields the corrected base orientation R_{base} :

$$R_{base} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} R_{base}^{raw} \quad (11)$$

The local relative rotation R_{loc} required by the gimbal chain is then computed by projecting the target orientation into the base frame via the transpose multiplication:

$$R_{loc} = R_{base}^T R_{tgt} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \quad (12)$$

2. *Pitch Solution* (θ_p): Due to the specific kinematic structure involving the tilt angle α , the element r_{22} of the local rotation matrix depends exclusively on the pitch angle. The pitch θ_p is thus derived directly from this element using structural constants K_1 and K_2 , which are functions of the tilt geometry:

$$\cos(\theta_p) = \frac{r_{22} - K_1}{K_2} \quad (13)$$

$$\theta_p = \text{atan2}\left(\sqrt{1 - \cos^2(\theta_p)}, \cos(\theta_p)\right) \quad (14)$$

where $K_1 \approx \sin^2(\alpha)$ and $K_2 \approx \cos^2(\alpha)$.

3. *Roll Solution* (θ_r): Once the pitch angle is established, the roll angle θ_r is calculated using the r_{12} and r_{32} elements of the matrix. This calculation includes a phase shift correction $\beta(\theta_p)$ that accounts for the projection of the canted axis:

$$\theta_r = \text{atan2}(r_{12}, r_{32}) - \beta(\theta_p) \quad (15)$$

4. *Yaw Solution* (θ_y): Finally, the yaw angle θ_y is isolated by "un-rotating" the local matrix by the computed roll and pitch angles. We calculate an intermediate matrix R_{rem} :

$$R_{rem} = R_{pitch}^{-1}(\theta_p) R_{roll}^{-1}(\theta_r) R_{loc} \quad (16)$$

The yaw value is then extracted using the arctangent of the remaining elements:

$$\theta_y = \text{atan2}(-R_{rem}[2, 0], R_{rem}[0, 0]) \quad (17)$$

This analytical approach ensures a deterministic and computationally efficient solution suitable for real-time control.

IV. CLOSED-LOOP CONTROL

The control architecture is designed as a hierarchical cascade, ensuring robust tracking performance while maintaining system safety. The architecture consists of three distinct layers: a discrete-event supervisor, a nonlinear motion controller (Backstepping), and a low-level current regulation loop (PI).

A. Discrete-Event Supervisor

To manage the operational modes and ensure safety against mechanical or kinematic failures, a finite state machine (FSM) is implemented as the high-level supervisor. The supervisor monitors system feedback—including joint positions q , currents I_{meas} , and operator commands—to determine the system's discrete state. The FSM is defined by four primary states:

1) *Idle* (*Status 0*): This is the default safe state. The actuators are disabled ($Enable = 0$), and the position command q_{cmd} is continuously updated to match the current position q_{curr} . This ensures that no sudden "jump" errors occur when the system transitions to active control.

2) *Run* (*Status 1*): The active tracking mode. The controller is enabled ($Enable = 1$), and the reference trajectory computed by the Inverse Kinematics (IK) module is passed directly to the motion controller. The supervisor continuously calculates the vector norm of the change in reference command, denoted as J_{mag} , to detect kinematic singularities.

3) *Fault* (*Status 2*): A latching error state triggered by critical safety violations. Transitions to this state occur immediately if the emergency stop (e_{stop}) is engaged or if the measured motor current exceeds a safety threshold ($|I_{meas}| > I_{max}$). In this state, actuation is disabled to protect the hardware.

4) *Mechanical Stop* (*Status 3*): A protection state designed to handle discontinuities in the IK solution (e.g., gimbal lock or quadrant wrapping). If the change in the requested joint angle exceeds a defined tolerance ($J_{mag} > J_{tol}$) within a single time step, the supervisor transitions to this state. Unlike the Fault state, the controller remains enabled (Active Braking), but the position command is locked to the last valid position (q_{lock}), effectively freezing the gimbal to prevent violent motion.

B. Nonlinear Backstepping Control

The core motion control is achieved using a Backstepping approach derived from the Lyapunov stability analysis. Considering the Lagrangian dynamics presented previously in (10), the control objective is to force the tracking error $e = q - q_d$ to zero.

The controller is formulated in two recursive steps. First, we define a virtual control reference ϕ (often denoted as $\dot{q}_r$) based on the position error:

$$\phi = \dot{q}_d - \Lambda e \quad (18)$$

where Λ is a positive definite gain matrix. This allows us to define a velocity tracking error surface z :

$$z = \dot{q} - \phi \quad (19)$$

To ensure global asymptotic stability, we select a Lyapunov function candidate incorporating the system's kinetic and potential energy characteristics:

$$V = \frac{1}{2} e^T e + \frac{1}{2} z^T M(q) z \quad (20)$$

Differentiating V with respect to time and substituting the dynamics from (10) yields the requisite control law. To cancel the inherent nonlinearities and enforce negative definiteness of $\dot{V}$, the final control torque τ is implemented as:

$$\tau = M(q)\dot{\phi} + C(q, \dot{q})\phi + G(q) - K_p e - K_d \dot{z} \quad (21)$$

In the practical implementation, the position error calculation e includes an angular wrapping function to handle the $-\pi$ to π crossover inherent in rotational joints. The terms K_p and K_d represent the proportional and derivative gains utilized to tune the convergence rate of the position and velocity errors, respectively.

C. Low-Level Current Control

The torque command τ generated by the backstepping controller acts as the setpoint for the low-level Brushless DC (BLDC) motor drivers. Since BLDC motor torque is directly proportional to current ($\tau \approx K_t \cdot I$), the control loop is closed via a Proportional-Integral (PI) current controller.

The desired current I_{cmd} is calculated from the desired torque using the motor torque constant K_t :

$$I_{cmd} = \frac{\tau_{des}}{K_t} \quad (22)$$

A PI regulator then minimizes the error between this request and the actual current sensed by the hardware (I_{meas}):

$$u_{PWM} = K_P(I_{cmd} - I_{meas}) + K_I \int (I_{cmd} - I_{meas})dt \quad (23)$$

This hierarchical structure ensures that the complex nonlinear dynamics are compensated for by the backstepping law, while the high-bandwidth current loop handles the electrical dynamics and provides the necessary torque to the physical plant.

V. RESULTS

To validate the proposed hierarchical control architecture, extensive simulations were conducted using the full nonlinear dynamic model. The performance was evaluated under three distinct operating conditions: (A) static stabilization with the base in an idle position, (B) dynamic trajectory tracking with a moving base, and (C) full system integration including the discrete-event supervisor.

A. Static Stabilization (Step Response)

In the first scenario, the gimbal base is held stationary, and the system is commanded to move from an initial condition to a fixed setpoint. This test isolates the controller's ability to regulate the plant dynamics without external base disturbances.

As presented in Fig. 1, the controller exhibits rapid and stable convergence. The joint trajectories (q_1, q_2, q_3) settle to their respective setpoints (q_d) within approximately 1.0 seconds. The pitch and yaw axes (Joints 2 and 3) converge smoothly to the reference trajectory, minimizing steady-state error. The roll axis (Joint 1) shows a negligible transient before settling.

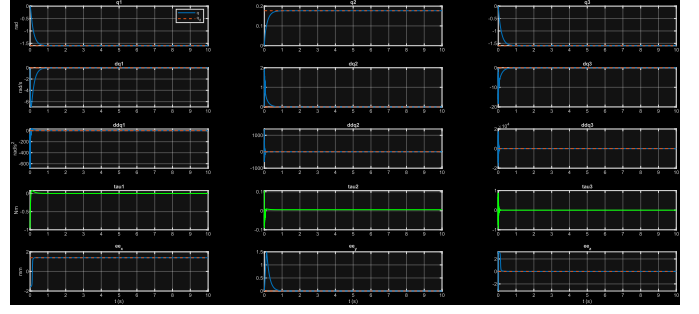


Fig. 1. Static stabilization response demonstrating rapid convergence of joint angles.

The end-effector error (ee), shown in the fourth row, converges asymptotically to zero, validating the effectiveness of the Lyapunov-based control law in driving the system to equilibrium. The control torques (τ) display a characteristic initial impulse to overcome system inertia, rapidly decaying to constant steady-state values required for gravity compensation.

B. Dynamic Trajectory Tracking

The second scenario evaluates the backstepping controller's robustness against external disturbances. The base is subjected to a continuous, time-varying position and orientation trajectory (depicted in the bottom panels of Fig. 2), simulating complex vehicle maneuvers.

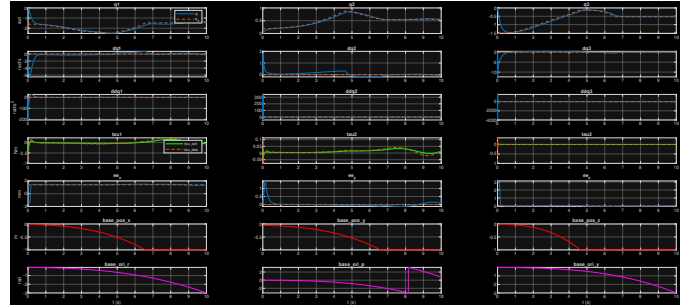


Fig. 2. Dynamic trajectory tracking under base excitation. The controller effectively rejects base disturbances.

Figure 2 illustrates the system's precise tracking capabilities under these dynamic conditions. Despite the significant base excitation, the actual joint angles (q , solid blue line) tightly track the desired trajectories (q_d , orange dashed line), resulting in minimal tracking error.

The smoothness of the velocity (dq) and acceleration (ddq) profiles indicates that the nonlinear control law effectively cancels the coupling effects, including Coriolis and centrifugal forces induced by the base motion. Furthermore, the control torques (τ) remain bounded and smooth, reflecting an efficient allocation of control effort without saturation or chattering in the continuous domain.

C. Full System Integration with Supervisor

Finally, the complete architecture is validated by integrating the discrete-event Stateflow supervisor with the continuous

plant dynamics. This scenario introduces hybrid system behavior, including discrete mode switching and signal quantization.

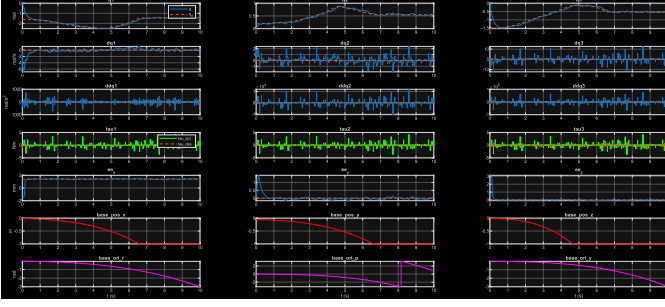


Fig. 3. Full system response with hierarchical supervision. Note the high-frequency switching characteristic of hybrid control.

The results in Fig. 3 demonstrate the robustness of the hybrid architecture. While the overall position tracking (q vs q_d) remains accurate, the inclusion of the discrete supervisor introduces high-frequency switching noise, particularly visible in the velocity estimates (dq) and the control torque signals (τ).

This phenomenon is a characteristic artifact of hybrid control systems where discrete logic gates interactions with continuous dynamics. Crucially, however, this high-frequency content does not destabilize the system or degrade the end-effector accuracy (ee), which remains bounded within acceptable tolerances. This confirms that the supervisory layer successfully enforces safety logic without compromising the stability of the low-level motion controller.

VI. EXPERIMENTAL SETUP

To verify the feasibility of the proposed control architecture, a custom experimental testbed was developed. The hardware realization focuses on a modular, lightweight design capable of executing the high-bandwidth nonlinear control algorithms in real-time through a distributed computing approach.

A. Mechanical Platform

The 3-axis gimbal structure was fabricated using additive manufacturing (3D printing). This construction method allowed for the rapid prototyping of complex geometries required to minimize the moment of inertia of the links while maintaining structural rigidity. The kinematic configuration corresponds directly to the model derived in Section III, ensuring that the software parameters match the physical plant.

As depicted in Fig. 4, the assembly is designed to allow a full range of motion for testing dynamic tracking and stabilization scenarios.

B. Electrical and Computing Architecture

The control system is implemented using a master-slave distributed architecture. The high-level computation is performed on a laptop, where the nonlinear control algorithms—including the Backstepping controller and the Inverse Kinematics solver—are implemented in C. This host computer



Fig. 4. Experimental setup of the 3-axis gimbal system.

is responsible for calculating the required joint torques based on the desired trajectory and system feedback. Additionally, a reverse kinematics module processes data from an MPU6050 Inertial Measurement Unit (IMU) mounted at the end-effector. This module estimates the base orientation in real-time, allowing the controller to compensate for external base disturbances effectively.

The computed control commands are transmitted via a high-speed communication interface to an ESP32 microcontroller, which serves as the low-level hardware abstraction layer. The ESP32 interfaces directly with the Electronic Speed Controllers (ESCs) to drive the three Brushless DC (BLDC) motors. It also aggregates sensor data from the high-resolution shaft encoders and motor current sensors, relaying this feedback loop back to the host computer. This separation of concerns allows the computationally intensive nonlinear control laws to run at high frequencies on the laptop while the ESP32 ensures precise, low-latency signal generation for the actuators.

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